

SOUTH EASTERN UNIVERSITY OF SRI LANKA
SECOND EXAMINATION IN BACHELOR OF INFORMATION AND
COMMUNICATION TECHNOLOGY - 2020/2021
SEMESTER – II, JANUARY 2024

CMS 22012 – Leadership & Communication Skills

Answer all Questions

Time: 02 hours.

1.
 - a. Briefly explain any five (05) importance of soft skills.
 - b. Briefly explain the techniques to gain Self – Confidence.
 - c. Briefly describe the advantages of Personal Accountability?
 - d. Explain the components of a presentation with suitable examples?

(04 * 25 = 100 Marks)
 (Total = 100 Marks)

2.
 - a. List out any of the five effective listening behavior. (10 Marks)
 - b. Briefly describe the importance of listening in a relationship. (20 Marks)
 - c. Briefly explain the Do's and Don'ts in a team (20 Marks)
 - d. Explain the step-by-step problem-solving approach in resolving a conflict. (20 Marks)
 - e. Explain the importance of Time Management with suitable examples. (20 Marks)

(Total = 100 Marks)

3.
 - a. Compare and contrast the verbal and non-verbal communication (20 Marks)
 - b. Briefly explain the following with suitable examples;
 - i. Passive communication
 - ii. Aggressive communication
 - iii. Assertive communication (3*15= 45 Marks)
 - c. "The ability to deliver an effective talk is one of the most valuable skills a leader can possess". What are the secrets of a successful public speaking? (15 Marks)
 - d. Briefly explain how to manage emotional intelligence? (20 Marks)

(Total = 100 Marks)

4.

a. Compare and contrast the following;

- i. Teamwork vs Working alone
- ii. Good leader vs team player

(3 * 10 = 20 Marks)

b. What are the reasons for an ineffective team meeting

(20 Marks)

c. "A team may be composed of people drawn from different functions, departments and disciplines who have been brought together for a specific project". Describe the steps in the development of workplace teams.

(20 Marks)

d. What are the Do's and Don'ts of Job Interviews?

(20 Marks)

e. Briefly explain the following with suitable examples;

- i. Effective ways to impress others.
- ii. First impression is the best impression.

(2 * 10 = 20 Marks)

(Total = 100 Marks)

**** END ****



SOUTH EASTERN UNIVERSITY OF SRI LANKA
DEPARTMENT OF INFORMATION & COMMUNICATION
TECHNOLOGY
SECOND EXAMINATION IN BICT - 2020/2021
SEMESTER II – JANUARY 2024
CIS22012 - DISTRIBUTED AND CLOUD COMPUTING

Answer ALL Questions

Time allowed: **TWO (02) Hours**

Question 01:

- a. Discuss the parameters of MPI_Send(), and MPI_Recv(). What is the purpose of additional parameter of MPI_Recv(). (05 Marks)
- b. Briefly Explain how a Synchronous mode of communication works. (04 Marks)
- c. Illustrate how an MPI works when 3 processes want to count a number of words in a paragraph for following scenario.
 - i. Using a common processor for calculating sum of counts.
 - ii. There is no common processor for calculating sum of counts. (08 Marks)
- d. Write MPI program to get the sum of two processes. The first process sends the number of 30 to the second process and the second process add that number with 70. (08 Marks)

[Total 25 Marks]

Question 02:

- a. Define how an IPC works for the following scenarios with suitable diagrams,
 - i. Ensure reliability.
 - ii. Ensure Ordering
 - iii. When the receiver side acknowledgment is missed.
 - iv. When the packet from sender is missing. (08 Marks)
- b. Briefly explain the event-based architecture using a real-life example. (04 Marks)
- c. Explain the principal work of BitTorrent with a suitable diagram. (05 Marks)
- d. Differentiate blocking and non-blocking primitive based on following scenarios.
 - i. TCP and UDP.
 - ii. Synchronous and Asynchronous.
 - iii. MPI_Ssend and MPI_Rsend.
 - iv. Printers and Websites. (08 Marks)

[Total 25 Marks]

Question 03:

- a. Provide real life examples for the following, (06 Marks)
- v. Internal Clock Synchronization
 - vi. External Clock Synchronization
 - vii. Idempotent
- b. Compare and contrast the following, (06 Marks)
- i. Centralized computing vs Parallel Computing
 - ii. MPI vs RMI
- c. A user wants to download a song called "Cause I-I-I'm in the stars tonight" from YouTube server. Explain the Transport Layer and Proxy layer process for the above scenario. (05 Marks)
- d. Write an RPC program for printing odd numbers from 0 to 10 on the client side.
- i. Server code

```
add_1_svc(numbers *argp, struct svc_req *rqstp)
{
    // place your code here
}
```

- ii. Client code

```
void
add_prog_1(char *host,int x,int y)
{
    CLIENT *clnt;
    int *result_1;
    numbers add_1_arg;

    result_1 = add_1(&add_1_arg, clnt);
    if(result_1 == (int *) NULL) {
        clnt_perror(clnt, "call failed");
    }
    // place your codes in suitable places
}
```

(08 Marks)

[Total 25 Marks]

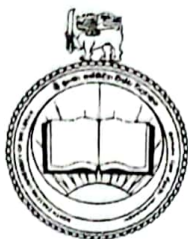
Question 04:

- a. Define the following terms in your own words, (06 Marks)
- i. RMI registry
 - ii. Naming.lookup()
 - iii. Naming.rebind()
- b. Differentiate the followings, (08 Marks)
- i. Garbage Collection vs. Distributed Garbage Collection
 - ii. Distributed Computing vs. Cloud Computing
 - iii. Direct Communication vs. Indirect Communication

- iv. SaaS vs. PaaS
- c. Based on your creativity, illustrate how a third wave of computer is applicable in medical domain.
(06 Marks)
- d. Write Linux commands for following RPC, RMI, and MPI functions.
 - i. Compile RMI file.
 - ii. Getting a port number for running RMI.
 - iii. Run a MPI program (Both commands are needed)
 - iv. Run RPC client file – Name of the client is “abc_client.c”

(05 Marks)

[Total 25 Marks]



SOUTH EASTERN UNIVERSITY OF SRI LANKA

**DEPARTMENT OF INFORMATION & COMMUNICATION
TECHNOLOGY**

SECOND EXAMINATION IN BICT - 2020/2021

SEMESTER II – JANUARY 2024

CIS22042 – PRACTICAL FOR DISTRIBUTED AND CLOUD COMPUTING

Answer **ALL** Questions

Time allowed: **THREE (03) Hours**

Instruction:

- Create a folder on the desktop, which should be named with your index number. E.g., ICTXXX
- Create a folder for each question (Q1, Q2, and Q3), and inside the folder, save all your work, including screenshots if necessary.
- Save your all files and folders from Ubuntu to Windows using the given shared folder.
- Save your work frequently.

Question 01: Remote Procedure Call (RPC)

- a. Develop an RPC program to calculate **area** and **perimeter** of a rectangle and triangle using two methods. (40 Marks)
- b. Design an RPC program where the client sends an array of integers to the server, and the server computes the **sum** of the elements in the array, returning the result to the client. (40 Marks)
- c. Q1 and Q2 outputs (Screenshots) (20 Marks)

[Total 100 Marks]

Question 02: Remote Method Invocation (RMI)

- a. Develop an RMI program to calculate the **area**, **perimeter**, and **volume** of a cylinder. (40 Marks)
- b. Create an RMI service to reverse a string. Develop a remote interface with a method that takes a string as input and returns its reverse. (40 Marks)
- c. Q1 and Q2 outputs (Screenshots) (20 Marks)

[Total 100 Marks]

Question 03: Message Passing Interface (MPI)

- a. Implement an MPI program for counting words in a book. The book consists of three (3) pages, and three (3) processes are counting the words simultaneously. Develop an MPI code for the above scenario.

Hint: The word count for each page will be sent by the users.

(40 Marks)

b. Factorial Calculation in Parallel. This activity involves four (4) processes:

- The first process acquires user input and forwards it to the second process.
- The second process receives the user input, adds it to the input received from the first process, and transfers the sum to the third process.
- The third process subtracts the received value from its user input and subsequently passes the result to the fourth process.
- The fourth process computes the factorial of the value received from the third process. If the result is negative, an error message is displayed.

(40 Marks)

c. Q1 and Q2 outputs (Screenshots)

(20 Marks)

[Total 100 Marks]

$$1 \times 2 \times 3 \times 4 \times 5$$
$$5 = \overset{2-1}{\underset{2}{(120)}}$$

SOUTH EASTERN UNIVERSITY OF SRI LANKA
SECOND EXAMINATION IN BACHELOR OF INFORMATION AND
COMMUNICATION TECHNOLOGY - 2020/2021
SEMESTER –II, JANUARY 2024
SWT 22012 – Internet Application Development

Index No:.....

Answer all Questions

Time Allowed: Two (02)Hours

Instructions

1. This paper consists of two (2) parts: **PART I** and **PART II**.
 2. **PART I** comprises **50** Multiple Choice Questions (MCQs). Answer all the MCQs.
 3. For each MCQ, choose the most appropriate answer from the options provided.
 4. Use only the provided marking sheet to mark your answers for the MCQ.
 5. Read the instructions on the marking sheet on how to mark the answer.
 6. **PART II** consists of two main structured essay questions. Write your answers for **PART II** in the provided booklet.
 7. Submit the *MCQ marking sheet* and *booklet separately* to the invigilator.
-

PART I

1. What does **URL** stand for?
 - a) Uniform Resource Locator
 - b) Universal Resource Locator
 - c) Unique Resource Locator
 - d) Unified Resource Locator
2. Which protocol is used for secure communication over a computer network?
 - a) FTP
 - b) HTTP
 - c) TCP/IP
 - d) HTTPS
3. What does **HTML** stand for?
 - a) Hyper Text Markup Language
 - b) Hyperlinks and Text Markup Language
 - c) High-level Text Markup Language
 - d) Hyper Transfer Markup Language

4. Which organization oversees the development of **web standards**?

a) W3C
b) IEEE

c) IETF
d) ISO
5. What is the primary function of a **web browser**?

a) File Management
b) Document Creation

c) Information Retrieval
d) Database Management
6. Which server primarily serves HTML documents and other resources in response to requests from **web browsers**?

a) Web Server
b) Application Server

c) Both
d) Neither
7. Which server executes **business logic**, interacts with databases, and generates dynamic content?

a) Web Server
b) Application Server

c) Both
d) Neither
8. Which of the following best describes **AJAX** in web development?

a) A programming language
b) A database management system
c) A technique for asynchronous data exchange between the client and server
d) An encryption protocol
9. Which server supports **EJBs (Enterprise Java Beans)** and other enterprise-level services?

a) Web Server
b) Application Server

c) Both
d) Neither
10. Which server hosts static content and handles **HTTP requests/responses**?

a) Web Server
b) Application Server

c) Both
d) Neither

11. Which server manages security, transactions, and clustering?

- a) Web Server
- b) Application Server
- c) Both
- d) Neither

12. Which design pattern focuses on creating objects based on a template of an existing object?

- a) Singleton
- b) Prototype
- c) Factory
- d) Decorator

13. Which design pattern separates object construction from its representation?

- a) Builder
- b) Singleton
- c) Factory Method
- d) Adapter

14. Which design pattern restricts the instantiation of a class to a single object?

- a) Singleton
- b) Decorator
- c) Observer
- d) Abstract Factory

15. Which programming language is primarily used for **client-side scripting** in web development?

- a) Python
- b) Java
- c) JavaScript
- d) Ruby

16. Which technology allows for **asynchronous communication** between the client and server?

- a) AJAX
- b) JSON
- c) XML
- d) REST

17. Which scripting language is often used for **server-side programming** and is embedded in **HTML documents**?

- a) JavaScript
- b) Python
- c) PHP
- d) Ruby

18. Which **protocol** defines how data is formatted and transmitted between client and server?
- a) HTTP
 - b) SMTP
 - c) FTP
 - d) UDP
19. Which protocol is commonly used for **Web Services**?
- a) HTTP
 - b) FTP
 - c) UDP
 - d) SMTP
20. Which technology is used for describing the functionalities provided by a **web service**?
- a) SOAP
 - b) JSON
 - c) XML
 - d) REST
21. Which type of web service relies on XML for message exchange?
- a) RESTful
 - b) JSON-RPC
 - c) SOAP
 - d) WSDL
22. Which approach to web services makes use of **HTTP methods** and is based on representing resources as URLs?
- a) RESTful
 - b) SOAP
 - c) WSDL
 - d) XML-RPC
23. Which type of database uses a tabular structure with rows and columns to organise data?
- a) NoSQL
 - b) Relational
 - c) Graph
 - d) Document-oriented
24. Which language is commonly used to interact with a **relational database**?
- a) HTML
 - b) SQL
 - c) JavaScript
 - d) XML
25. Which **JavaScript framework** is used for building user interfaces and single-page applications?
- a) Angular
 - b) React
 - c) Vue.js
 - d) Ember.js

26. Which **front-end framework** was developed by Google?

- a) Angular
- b) React
- c) Vue.js
- d) Ember.js

27. Which front-end framework follows the **component-based architecture**?

- a) Angular
- b) React
- c) Vue.js
- d) Ember.js

28. Which front-end framework uses a **virtual DOM** for better performance?

- a) Angular
- b) React
- c) Vue.js
- d) Ember.js

29. Which programming language is commonly associated with the **Node.js framework**?

- a) Python
- b) JavaScript
- c) Ruby
- d) PHP

30. Which back-end framework is based on **Python**?

- a) Express.js
- b) Ruby on Rails
- c) Spring Boot
- d) Django

31. Which framework is primarily used for building user interfaces in **JavaScript**?

- a) Angular
- b) Django
- c) Flask
- d) Laravel

32. Which **back-end framework** is used for building scalable network applications in JavaScript?

- a) Django
- b) Flask
- c) Express.js
- d) Ruby on Rails

33. Which **back-end framework** is based on the **Ruby** programming language?

- a) Django
- b) Flask
- c) Express.js
- d) Ruby on Rails

34. What is the primary goal of **Software Configuration Management (SCM)**?
- a) Manage software development processes
 - b) Ensure the security of software systems
 - c) Monitor server uptime
 - d) Optimise database performance
35. Which **Software Configuration Management (SCM)** system is designed for distributed version control?
- a) Git
 - b) Subversion (SVN)
 - c) Mercurial
 - d) Concurrent Versions System (CVS)
36. Which **Software Configuration Management (SCM)** system is centralised, with a single server storing the version history?
- a) Git
 - b) Subversion (SVN)
 - c) Mercurial
 - d) Concurrent Versions System (CVS)
37. Which **Software Configuration Management (SCM)** system uses changesets as its fundamental versioning unit?
- a) Git
 - b) Subversion (SVN)
 - c) Mercurial
 - d) Concurrent Versions System (CVS)
38. Which **Software Configuration Management (SCM)** system was developed by **Linus Torvalds**?
- a) Git
 - b) Subversion (SVN)
 - c) Mercurial
 - d) Concurrent Versions System (CVS)

39. Which **Software Configuration Management (SCM)** system is known for its speed and efficiency in handling branching and merging?
- a) Git
 - b) Subversion (SVN)
 - c) Mercurial
 - d) Concurrent Versions System (CVS)
40. Which term refers to the process of converting data into a format suitable for transmission over the Internet?
- a) Serialisation
 - b) Decryption
 - c) Compression
 - d) Encryption
41. Which protocol is commonly used for **Web Services**?
- a) HTTP
 - b) FTP
 - c) UDP
 - d) SMTP
42. Which style of web service design emphasises **minimalism and human-readable URLs**?
- a) SOAP
 - b) REST
 - c) XML-RPC
 - d) WSDL
43. Which type of web service relies on **XML** for message exchange?
- a) RESTful
 - b) JSON-RPC
 - c) SOAP
 - d) WSDL
44. Which language is commonly used to interact with a relational database?
- a) HTML
 - b) SQL
 - c) JavaScript
 - d) XML
45. Which **Software Configuration Management (SCM)** system was developed by Linus Torvalds?
- a) Git
 - b) Subversion (SVN)
 - c) Mercurial
 - d) CVS

46. Which architecture separates the user interface from the business logic and data storage?

- a) Monolithic architecture
- b) Microservices architecture
- c) MVC architecture
- d) Serverless architecture

47. Which feature of **Software Configuration Management (SCM)** allows developers to work on a local copy of a repository and later sync changes with a central repository?

- a) Branching
- b) Merging
- c) Committing
- d) Cloning

48. What is the purpose of a "**Merge**" operation in version control systems like **Git** or **SVN**?

- a) To create a backup of the repository
- b) To combine changes from different branches or source
- c) To delete unwanted commits
- d) To revert the repository to a previous state

49. What does the term "**Baseline**" signify in **Software Configuration Management (SCM)**?

- a) It refers to the final stage of the software development lifecycle
- b) It is a snapshot of a configuration item that serves as a reference point.
- c) It is a dynamic process to continuously update software features.
- d) It represents the initial design phase of a software project.

50. What does the acronym "**CI/CD**" stand for in the context **Software Configuration Management (SCM)**?

- a) Continuous Integration/Continuous Deployment
- b) Controlled Iteration/Continuous Development
- c) Centralised Infrastructure/Continuous Deployment
- d) Continuous Improvement/Continuous Deployment

[01 x 50 = 50 Marks]

PART II

Question 01:

- a. Explain the fundamental differences between a **web server** and an **application server** in the context of internet (web) application development. (05 Marks)
- b. Compare and contrast **client-side** and **server-side programming** in web applications, highlighting their respective roles and functionalities. (05 Marks)
- c. Describe the impact of **front-end frameworks** on modern web development practices (05 Marks)
- d. Discuss the importance of **back-end frameworks** in web development and provide examples of how they streamline development processes and enhance application scalability. (10 Marks)

[25 Marks]

Question 02:

- a. Define what is **web services** and state the **purpose of web services** in software development. (05 Marks)
- b. What are the primary types of **web services**? And how do they differ in their communication protocols? (05 Marks)
- c. Explain the role and significance of "**Software Configuration Management (SCM)**" in the software development lifecycle, emphasising its impact on version control, team collaboration, and project scalability. (05 Marks)
- d. Write Short notes for the following:
 - I. Model-View-Controller (MVC)
 - II. Web API
 - III. HTML5
 - IV. Configuration Management Plan (CMP)(10 Marks)

[25 Marks]

[Total: 100 Marks]

**** End of the Paper ****

SOUTH EASTERN UNIVERSITY OF SRI LANKA
 SECOND EXAMINATION IN BACHELOR OF INFORMATION AND
 COMMUNICATION TECHNOLOGY - 2020/2021
 SEMESTER – II, JANUARY/ FEBRUARY 2024
 UCT 22022– PRACTICAL FOR MICRO CONTROLLER SYSTEMS
 PROGRAMMING

Answer all Questions

Time Allowed: 03 Hours.

- Use Proteus and Arduino IDE to accomplish the task given below.
- You are requested to submit a **PDF document** having the circuit diagram and the Arduino Program attached to it. Also, attach necessary screenshots (diagrams output) and simulation images with the necessary INFORMATION.
- **Rename your PDF as Your Registration Number**

Question 01

- Create a calculator using Arduino, incorporating a 4x4 keypad (specifically the calculator keypad from Proteus) and a 16x2 Alphanumeric Liquid Crystal Display. This calculator should be capable of performing fundamental mathematical operations, such as addition, subtraction, multiplication, and division, using two whole numbers. Utilize an LCD that interfaces with Arduino to serve as the calculator's screen.
 - multiply your index number digits by 2 and display your outputs

Below are some images of the result

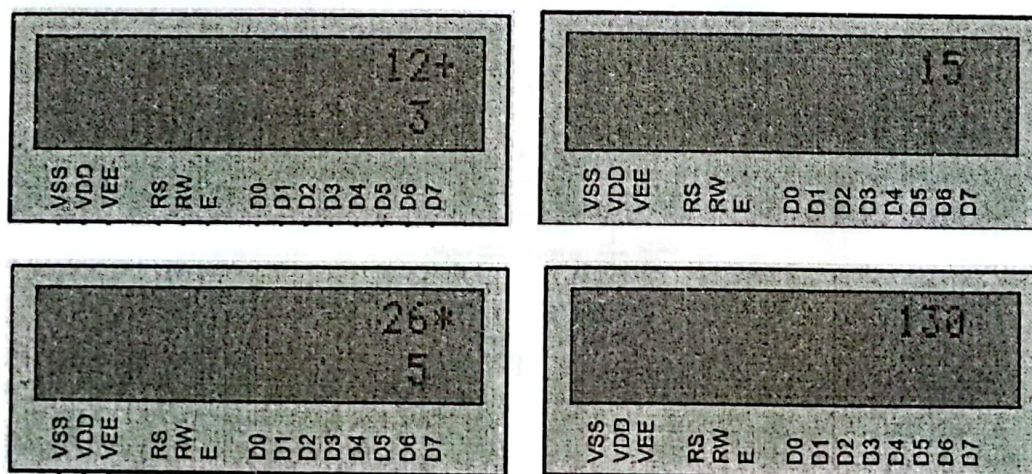


Figure 01: Results of the Electronic Calculator

[40 Marks]

Question 02

Create a circuit simulation featuring **eight (08) LEDs**, all connected to Port C of a PIC16F877A microcontroller. The microcontroller should be driven by an 8MHz crystal oscillator. Your program should be designed to alternate the LEDs, with each LED turning on with a one-second delay. The result of the circuit will be a visually engaging display, as the LEDs appear to chase one another in sequence.

The following information is useful for your tasks:

PIC16F877A, 8 LEDs, 8 290 Ohm resistors, 5V power supply, 2 22pF Capacitors and 1 Crystal Oscillator

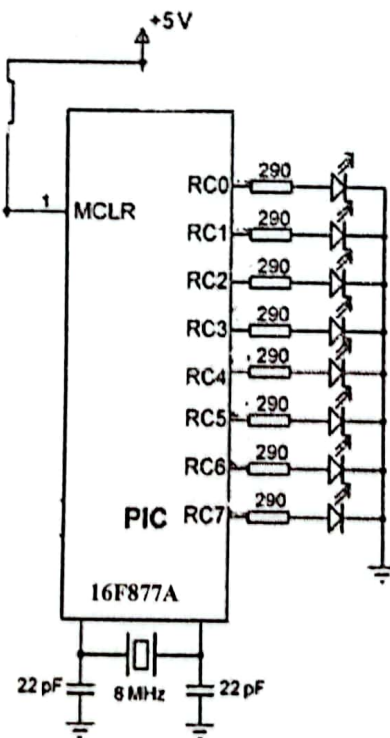


Figure 02: PIC diagram

[30 Marks]

Question 03

Design a circuit using a PIR sensor, a Servo motor, and a DC motor to make a simple Radar system with Arduino Interrupt. In the system, the servo should rotate through 180 degrees from left to right and right to left. Further, the DC motor connected to the Arduino should continuously rotate until an interrupt occurs. Whenever the PIR sensor detects a motion, the DC motor should stop running and an LED and buzzer should turn on.

[30 Marks]

[Total 100 Marks]

SOUTH EASTERN UNIVERSITY OF SRI LANKA
SECOND EXAMINATION IN BACHELOR OF INFORMATION AND
COMMUNICATION TECHNOLOGY - 2020/2021
SEMESTER – II, JANUARY/ FEBRUARY 2024

UCT 22011– MICRO CONTROLLER SYSTEMS PROGRAMMING

Answer all Questions

Time Allowed: 01 hour.

Question 01

- a. What is an Interrupt? Explain with examples.

(26 Marks)

- b. What is a microcontroller? What is a microprocessor? Explain the main differences between a microprocessor and a microcontroller.

(24 Marks)

[50 Marks]

Question 02

- a. Describe the major six (06) features taken into account on choosing a microcontroller for an application/project.

(26 Marks)

- b. Define and differentiate the following terms

- i. Random Access Memory Vs Programmable Read-Only Memory
- ii. Erasable Programmable Read-Only Memory Vs Electrically Erasable Programmable Read-Only Memory
- iii. RISC Vs CISC

(24 Marks)

[50 Marks]

[Total 100 Marks]

SOUTH EASTERN UNIVERSITY OF SRI LANKA**SECOND EXAMINATION IN BACHELOR OF INFORMATION AND
COMMUNICATION TECHNOLOGY - 2020/2021****SEMESTER – II, JANUARY - 2024****CIS22032 – E-Commerce Strategies and Architecture****Answer all (04) questions.****Time allowed: 02 hours**

Question 01:

- a) **Describe** the term E-Commerce and **differentiate** the term E-Commerce from E-Business.

(20 Marks)

- b) Briefly describe **five (5)** unique features of E-Commerce technology.

(25 Marks)

- c) **Describe** at least **two (2)** benefits of E-Commerce in comparison to traditional commerce considering both organizational and customer perspectives.

(20 Marks)

- d) Discuss the technical and non-technical limitations of E-Commerce.

(20 Marks)

- e) Describe each of the following types of E-Commerce and distinguish among them.

- i. Business-to-Consumer (B2C)
- ii. Business-to-Business (B2B)
- iii. Consumer-to-Consumer (C2C)

(15 Marks)

[100 Marks]

Question 02:

- a) Briefly describe at least **five (5)** factors to consider when deciding on an E-Commerce business model.

(25 Marks)

- b) Describe the following business-to-consumer business models with examples and identify their revenue model(s).

- i. E-tailer
- ii. Community Provider
- iii. Content Provider
- iv. Transaction Broker
- v. Service Provider

(35 Marks)

- c) Describe the **five (5)** major types of revenue models and provide at least one (1) example for each of them.

(30 Marks)

- d) Identify the revenue model of the business scenario given below.

- i. A company operates an online store where customers can purchase clothing, electronics, and other products. Customers pay for each item they select, and the company ships the products to them.
- ii. A platform offers unlimited access to a library of movies, TV shows, and documentaries for a monthly fee. Users can watch as much content as they want as long as they maintain their subscriptions.
- iii. A social networking site allows users to create profiles, connect with friends, and share updates. The platform displays targeted advertisements to users based on their interests and online behaviour.

- iv. A blogger writes product reviews on a website and includes affiliate links in the content. The blogger earns a commission when readers click on these links and make purchases.

(10 Marks)

[100 Marks]

Question 03:

- a) Briefly describe the steps involved in creating an E-Commerce site.

(20 Marks)

- b) **Explain** the multi-tier E-Commerce architecture and **briefly describe** the functionalities provided by web servers.

(30 Marks)

- c) Analyse the factors that are most important for creating a successful E-Commerce presence and the factors that annoy the users of a site.

(20 Marks)

- d) **Identify** the factors that affect the speed, capacity, and scalability of an e-commerce site and **discuss** the ways/ techniques to scale the hardware needs of an e-commerce site.

(10 Marks)

- e) Identify the major considerations involved in choosing Web server and E-Commerce merchant server software.

(20 Marks)

[100 Marks]

Question 04:

a) Describe each of the following terms.

- i. Internet
- ii. Web

(20 Marks)

b) Discuss the rise of e-commerce after the introduction of Web 2.0 with the help of the features and services of Web 2.0.

(30 Marks)

c) List at least **five (5)** names of organizations that influence the Internet, and monitor its operations. Briefly state the operations performed by each organization.

(10 Marks)

d) **Identify and describe** the limitations of the current Internet and **propose** your solutions to overcome them.

(20 Marks)

e) **Discuss** how traditional marketing and E-Commerce differ in terms of the privacy of their customers and **suggest** ways to protect the privacy of customers on the Internet.

(20 Marks)

[100 Marks]

[Total 400 Marks]

****END****

SOUTH EASTERN UNIVERSITY OF SRI LANKA
SECOND EXAMINATION IN BACHELOR OF INFORMATION AND
COMMUNICATION TECHNOLOGY - 2020/2021
SEMESTER – II, JANUARY 2024
CIS 22022 – Information Assurance and Forensics

Answer ALL (04) Questions.

Time allowed: 02 hours

Question 01:

- A) Draw a suitable 3D model for Information Assurance (IA) by including the five pillars of IA, states of the information, and security countermeasures.

(04 marks)

- B) State and briefly describe the three (03) main principles of Information Security.

(06 marks)

- C) Consider the use of **Asymmetric Cryptography** in public key encryption and Digital signature and fill following blanks using the appropriate option given on the bracket.

-⁽ⁱ⁾ (sender/receiver)'s⁽ⁱⁱ⁾ (public/private) key can be used to **encrypt a message**.
-⁽ⁱⁱⁱ⁾ (sender/receiver)'s^(iv) (public/private) key can be used to **decrypt a ciphertext**.
-^(v) (sender/receiver)'s^(vi) (public/private) key can be used to **sign a Digital signature**.
-^(vii) (sender/receiver)'s^(viii) (public/private) key can be used to **verify a Digital signature**.

(06 marks)

- D) Find the encrypted cipher text for the sentence “**this is the password**” using the **Polyalphabetic Substitution** technique by following the given instructions below.

- i. You need to create two reference tables for odd position and even position.

- ii. Use Caesar cipher with key as 4 to prepare an odd position table.
[Rule for Caesar cipher is $C_i = E(P_i)$ where $i_c \Rightarrow (i + key)_p$]
- iii. Use the word “dict” as the key to prepare an even position table.
[use the key at the *beginning* of the table]
- iv. Write the encrypted cipher text using the prepared reference tables.

(09 marks)

[25 marks]

Question 02:

- A) What is Digital forensics? What is meant by “Seizure” in digital forensics?

(08 marks)

- B) State and briefly describe the components of the Investigation triad in digital forensics.

(06 marks)

- C) State Five (05) main rules of electronic evidence collection.

(05 marks)

- D) Write a short note on the following legal personals/ processes in evidence collection.

- i. Digital Evidence First Responder (DEFR)
- ii. Digital Evidence Specialist (DES)
- iii. Affidavit

(06 marks)

[25 marks]

Question 03:

- A) Define the data owner, data custodian, and data user of an organization. Differentiate their responsibilities on the organization’s data.

(07 marks)

B) *"Organizations are focusing on building security into the SDLC, to provide security assurance by preventing security problems before they begin". State five (05) software development security problems that could lead to security threats.*

(05 marks)

C) **Choose** the correct answer from the given options on the special properties relevant to the Hash function in information security.

- i. Output hash size for messages with different input sizes (Same/ Different)
- ii. Usage of key to create and verify hash (Keyed/ Not keyed)
- iii. The ability to calculate the original message from its output hash (Feasible/ Infeasible)
- iv. The ability to create the same hash output for different inputs (Feasible/ Infeasible)
- v. The scope and availability of access (Public / Private)

(05 marks)

D) *"The threat from external sources increases when an organization connects to the Internet". Write a short note on the following threats/ attacks related to information security.*

- i. Espionage or Trespass
- ii. Distributed Denial-of-Service (DDoS)
- iii. Man-in-the-Middle
- iv. Bot attack

(08 marks)

[25 marks]

Question 04:

A) **State and define four (04) concepts to build a secure and levels of controlled access environment in information systems.**

(08 marks)

B) **Write down the five (05) criteria to be followed to enforce security policy in an organization effectively and legally.**

(05 marks)

C) **Briefly discuss** the importance of ethics and education on information security practices.

(06 marks)

D) Write a **short note** on the following security perimeters related to information security.

- i. Firewall
- ii. Demilitarized Zones
- iii. Proxy Servers

(06 marks)

*** END OF PAPER ***